ANUDIP FOUNDATION · DBA_AI · ANP-D7635

AIRLINE ROUTE & FLIGHT OPERATIONS ANALYTICS – INDIGO

An academic data analytics project using MySQL validation and Power BI reporting.

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PROJECT SCOPE

What the project analyzes



Scheduled records

Scheduled flight records associated with IndiGo.



Route diversity

Route diversity, airport activity, scheduled duration, and departure-hour distribution.



Analytics workflow

MySQL validates and summarizes the data; Power BI turns the analysis into interactive dashboards.

This is a dataset-based study, not a live connection to airline operational systems.

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METHOD

Objectives and analytical workflow

01

Inspect

Inspect and validate the flight dataset.

02

Analyze

Analyze records, routes, airports, durations, and departure hours with SQL.

03

Build

Build interactive Power BI report pages with KPIs, charts, slicers, and calculated fields.

04

Present

Present evidence-based findings and limitations.

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DATASET

1,049

flight records

Indigo_Fleet_Working.csv

Database: indigo_database

Table: flight_data

FIELDS & TOOLS

Dataset and technology stack



Fields

ORIGIN, DEST, CRS_DEP_DATETIME, CRS_ARR_DATETIME.



Tools

MySQL, SQL, Power BI Desktop, Power Query, DAX, CSV/Excel

Project repository on GitHub

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DATA VALIDATION

Validation established a reliable analysis base

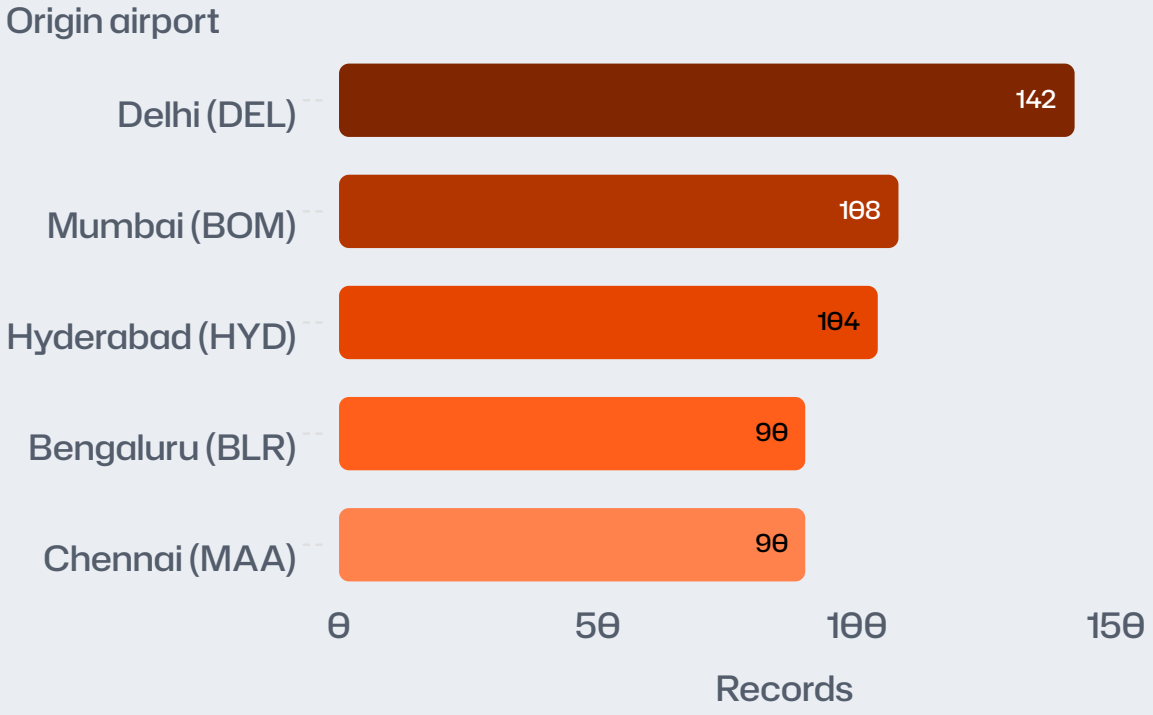
Validation check	Result
Records	1,049 total records.
Route combinations	432 distinct origin–destination route combinations.
Airports	58 distinct origin airports and 58 distinct destination airports.
Missing values	No missing values in ORIGIN, DEST, CRS_DEP_DATETIME, or CRS_ARR_DATETIME.
Duplicate composite keys	No duplicate composite keys across origin, destination, scheduled departure, and scheduled arrival.
Scheduled duration	All 1,049 records had positive scheduled flight durations.

Scheduled duration was calculated from the difference between scheduled arrival and departure timestamps.

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SQL FINDINGS

SQL analysis highlights the busiest records and routes



EXAMPLE HIGH-FREQUENCY ROUTES

Route	Count
Mumbai → Delhi	17
Delhi → Mumbai	15
Kolkata → Delhi	12
Chennai → Hyderabad	11
Delhi → Kolkata	11

These are counts in the supplied dataset, not claims about real-world flight frequency.

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POWER BI · REPORT PAGE 1

Power BI Executive Overview

The first report page summarizes scale and distribution through KPI cards for records, routes, origin airports, and destination airports, plus route, origin, destination, and departure-hour visuals.

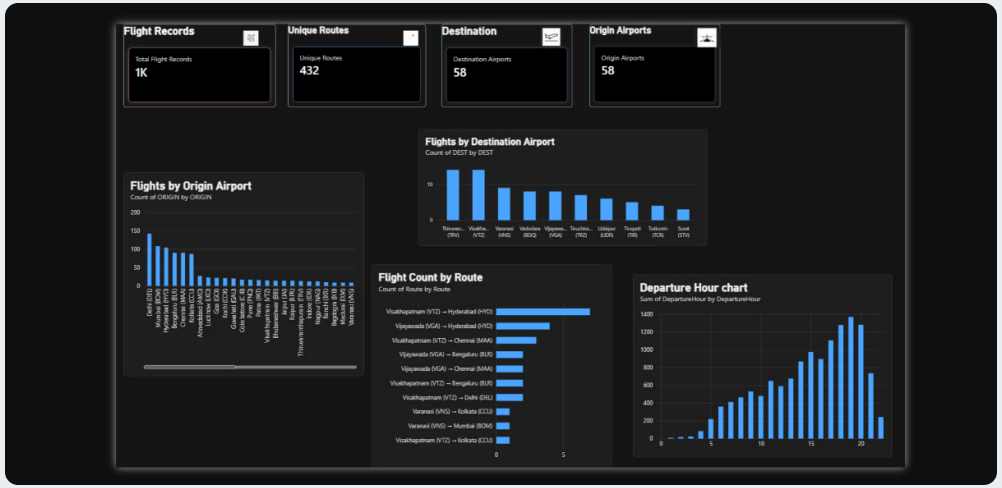


Figure 1: Power BI Executive Overview Dashboard

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POWER BI · REPORT PAGE 2

Route and Flight Analysis supports exploration

The second page compares average scheduled duration by route and provides origin and destination airport slicers so viewers can explore selected routes and airport combinations.

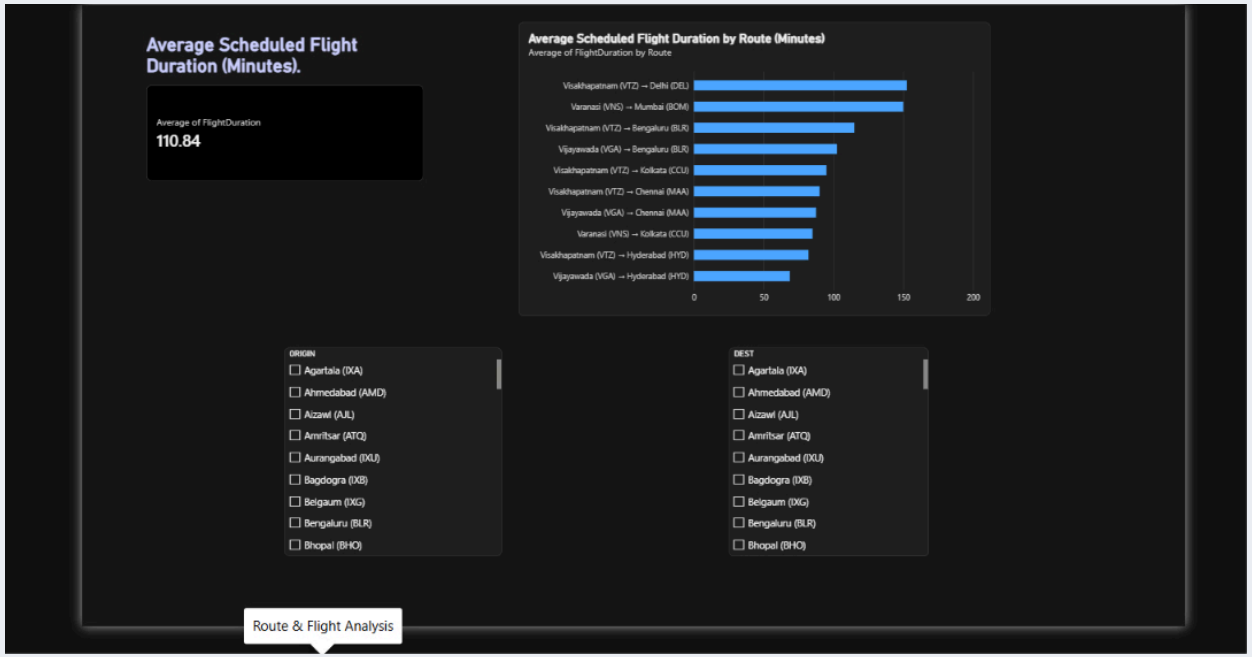


Figure 2: Power BI Route & Flight Analysis Dashboard

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POWER BI · REPORT PAGE 3

Flight Operations & Performance shows scheduled patterns

The third page combines origin distribution, route volume, scheduled flights by departure hour, average scheduled duration, and route detail into one visual exploration.



Figure 3: Power BI Flight Operations & Performance Dashboard

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POWER BI CALCULATIONS

Power BI calculations turn timestamps into insight

01

FlightDuration

Scheduled arrival minus scheduled departure, expressed in minutes.

02

Route

A combined origin and destination airport label.

03

DepartureHour

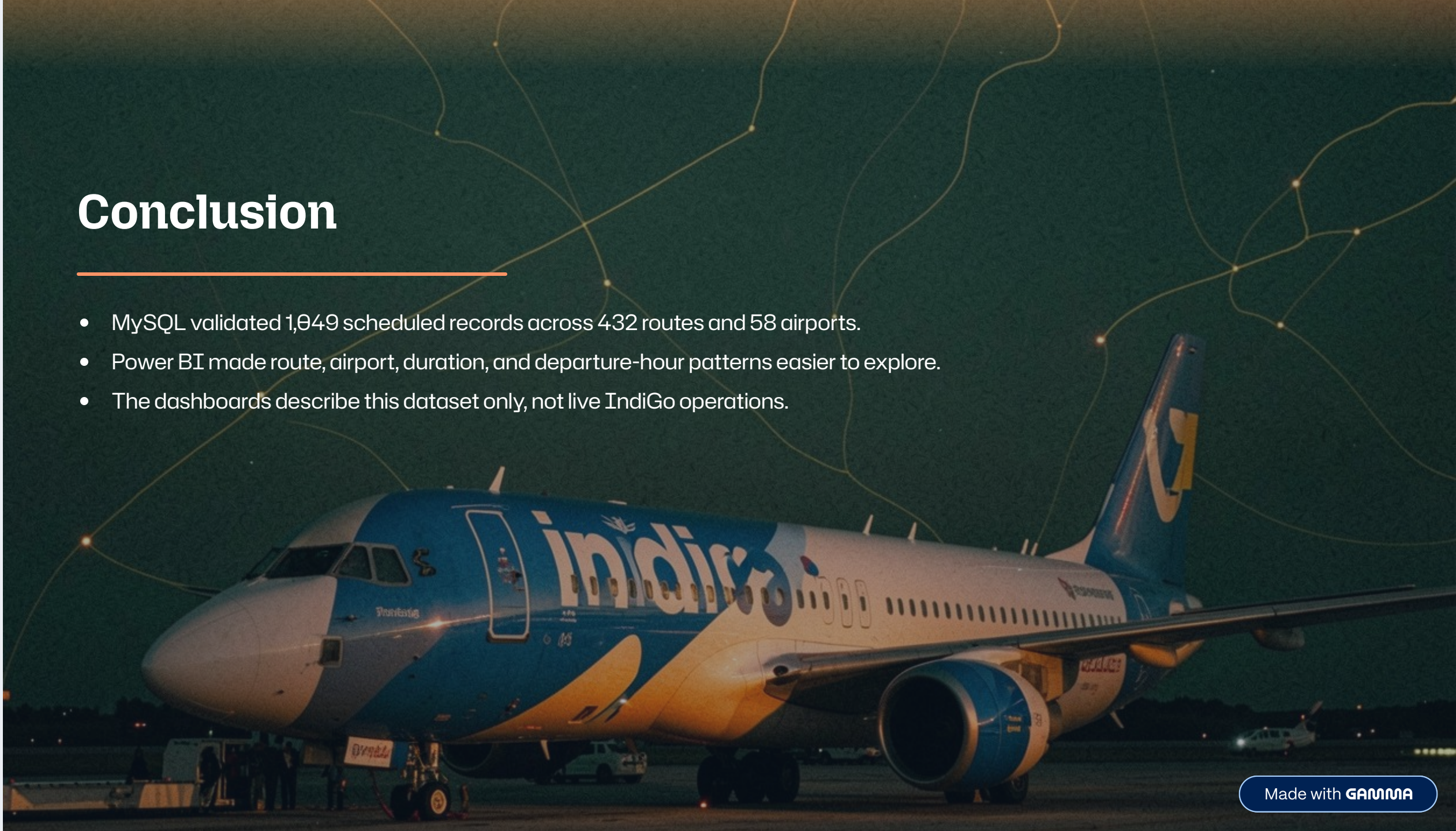
The scheduled departure hour extracted from CRS_DEP_DATETIME.

These calculated fields support route, duration, and time-of-day visuals.

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Conclusion

- MySQL validated 1,049 scheduled records across 432 routes and 58 airports.
- Power BI made route, airport, duration, and departure-hour patterns easier to explore.
- The dashboards describe this dataset only, not live IndiGo operations.



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